50:1 Micro Metal Gearmotor HPCB 6V with Extended Motor Shaft

Pololu item #: 3073 **141** in stock
Brand: [Pololu] [supply outlook]
Status: Active and Preferred
✓RoHS 3
Free add-on shipping in USA
Free shipping in USA over $75

Price break	Unit price (US$)
1	23.45
5	21.57
25	19.85
100	18.26

Quantity: 1 Add to cart
backorders *allowed* Add to wish list

This is a miniature brushed DC metal gearmotor with a gearbox cross section of 10×12 mm and a 9 mm long, 3 mm diameter D-shaped gearbox output shaft.

motor/brush type	gearbox	encoder
HPCB 6V: high-power 6V with carbon brushes	51.45:1 with brass plates	encoder-compatible (extended motor shaft)

voltage	no-load performance	stall extrapolation*
6 V	650 RPM, 150 mA	0.74 kg·cm (10 oz·in), 1.5 A

* Note: Stall torque and stall current specifications are theoretical values; stalls could damage the motor or gearbox.

Alternatives available with variations in these parameter(s): gear ratio motor type encoder **Select variant…**

or or .

Description **Specs (17)** **Pictures (17)** **Resources (3)** **FAQs (2)** **On the blog (3)** **Distributors (47)**

Overview

Our Micro Metal Gearmotor family consists of tiny brushed DC metal gearmotors with nitride-hardened martensitic stainless steel gears in a wide range of gear ratios, motor windings, brushes, and encoder configurations:

Micro Metal Gearmotor next to a US quarter dollar for size reference.

Gear ratio options	Motor winding/brush options	Encoder options
• 5:1 • 10:1 • 15:1 • 30:1 • 50:1 • 75:1 • 100:1 • 150:1 • 210:1 • 250:1 • 298:1 • 380:1 • 1000:1	• **HPCB 12V:** high-power 12V with long-life carbon brushes • **HPCB 6V:** high-power 6V with long-life carbon brushes • **HP 6V:** high-power 6V with precious metal brushes • **MP 6V:** medium-power 6V with precious metal brushes • **LP 6V:** low-power 6V with precious metal brushes	• integrated 12 CPR quadrature encoder with back connector • integrated 12 CPR quadrature encoder with side connector • encoder-compatible (extended motor shaft for adding an encoder) • no encoder

The following comparison table shows basic specifications for all of the available versions (see the **Micro Metal Gearmotor datasheet** (5MB pdf) for additional information, including detailed performance graphs):

Rated Voltage	Stall Current	No-Load Current	No-Load Speed (RPM)	Extrapolated Stall Torque (kg·cm)	Extrapolated Stall Torque (oz·in)	Max Power (W)	Approx Gear Ratio	 No Encoder	 w/ Extended Motor Shaft	 w/ Encoder, Back Conn.	 w/ Encoder, Side Conn.
HPCB 12V (high-power, carbon brushes)											
12 V	0.75 A	100 mA	6800	0.09	1.3	–	5:1	#3036	#3047	#5204	#5205
		80 mA	3400	0.17	2.4	1.5	10:1	#3037	#3048	#5206	#5207
			2200	0.25	3.5	1.4	15:1	#4788	#4789	#5208	#5209
			1100	0.39	5.4	1.1	30:1	#3038	#3049	#5210	#5211
			650	0.67	9.3	1.1	50:1	#3039	#3050	#5212	#5213
			450	1.0	14	1.1	75:1	#3040	#3051	#5214	#5215
			330	1.3	18	1.1	100:1	#3041	#3052	#5216	#5217
			220	1.8	25	1.0	150:1	#3042	#3053	#5218	#5219
			160	2.5	35	1.0	210:1	#3043	#3054	#5220	#5221
			130	3.0	42	1.1	250:1	#3044	#3055	#5222	#5223
			110	3.3	46	1.0	298:1	#3045	#3056	#5224	#5225
			85	5.0	69	1.1	380:1	#4798	#4799	#5226	#5227
			35	10	140	–	1000:1	#3046	#3057	#5228	#5229

Rated Voltage	Stall Current	No-Load Current	No-Load Speed (RPM)	Extrapolated Stall Torque (kg·cm)	Extrapolated Stall Torque (oz·in)	Max Power (W)	Approx Gear Ratio	No Encoder	w/ Extended Motor Shaft	w/ Encoder, Back Conn.	w/ Encoder, Side Conn.
HPCB 6V (high-power, carbon brushes)											
6 V	1.5 A	170 mA	6500	0.09	1.3	–	5:1	#3060	#3082	#5178	#5179
		150 mA	3300	0.17	2.4	1.3	10:1	#3061	#3071	#5180	#5181
			2100	0.25	3.5	1.3	15:1	#4786	#4787	#5182	#5183
			1100	0.45	6.2	1.2	30:1	#3062	#3072	#5184	#5185
			650	0.74	10	1.2	50:1	#3063	#3073	#5186	#5187
			430	1.1	15	1.3	75:1	#3064	#3074	#5188	#5189
			330	1.6	22	1.3	100:1	#3065	#3075	#5190	#5191
			220	2.0	28	1.1	150:1	#3066	#3076	#5192	#5193
			160	2.8	39	1.1	210:1	#3067	#3077	#5194	#5195
			130	3.2	44	1.1	250:1	#3068	#3078	#5196	#5197
			110	3.4	47	1.0	298:1	#3069	#3079	#5198	#5199
			85	5.0	69	1.1	380:1	#4796	#4797	#5200	#5201
			33	11	150	–	1000:1	#3070	#3080	#5202	#5203

Rated Voltage	Stall Current	No-Load Current	No-Load Speed (RPM)	Extrapolated Stall Torque (kg·cm)	Extrapolated Stall Torque (oz·in)	Max Power (W)	Approx Gear Ratio	No Encoder	w/ Extended Motor Shaft	w/ Encoder, Back Conn.	w/ Encoder, Side Conn.
HP 6V (high-power)											
6 V	1.6 A	120 mA	6100	0.11	1.5	–	5:1	#1000	#2210	#5152	#5153
		100 mA	3100	0.22	3.0	1.6	10:1	#999	#2211	#5154	#5155
			2000	0.30	4.2	1.5	15:1	#4784	#4785	#5156	#5157
			1000	0.57	7.9	1.5	30:1	#1093	#2212	#5158	#5159
			590	0.86	12	1.3	50:1	#998	#2213	#5160	#5161
			410	1.3	18	1.4	75:1	#2361	#2215	#5162	#5163
			310	1.7	24	1.3	100:1	#1101	#2214	#5164	#5165
			210	2.4	33	1.2	150:1	#997	#2386	#5166	#5167
			150	3.0	42	1.1	210:1	#996	#2216	#5168	#5169

			120	3.4	47	1.1	250:1	#995	#2217	#5170	#5171
			100	4.0	56	1.1	298:1	#994	#2218	#5172	#5173
			84	5.5	76	1.1	380:1	#4794	#4795	#5174	#5175
			31	12	170	–	1000:1	#1595	#2373	#5176	#5177

Rated Voltage	Stall Current	No-Load Current	No-Load Speed (RPM)	Extrapolated Stall Torque		Max Power (W)	Approx Gear Ratio	No Encoder	w/ Extended Motor Shaft	w/ Encoder, Back Conn.	w/ Encoder, Side Conn.
				(kg·cm)	(oz·in)						
MP 6V (medium-power)											
6 V	0.67 A	80 mA	4400	0.06	0.8	–	5:1	#2362	#2376	#5126	#5127
		70 mA	2200	0.11	1.5	–	10:1	#2363	#2377	#5128	#5129
			1400	0.20	2.8	0.70	15:1	#4782	#4783	#5130	#5131
			720	0.33	4.6	0.57	30:1	#2364	#2378	#5132	#5133
			420	0.54	7.5	0.55	50:1	#2365	#2379	#5134	#5135
			290	0.78	11	0.54	75:1	#2366	#2380	#5136	#5137
			220	0.94	13	0.50	100:1	#2367	#2381	#5138	#5139
			150	1.3	18	0.48	150:1	#2368	#2382	#5140	#5141
			100	1.7	24	0.46	210:1	#2369	#2383	#5142	#5143
			88	2.2	31	0.48	250:1	#2370	#2384	#5144	#5145
			73	2.4	33	0.44	298:1	#2371	#2385	#5146	#5147
			57	3.6	50	0.53	380:1	#4792	#4793	#5148	#5149
			22	6.7	93	–	1000:1	#2372	#3059	#5150	#5151

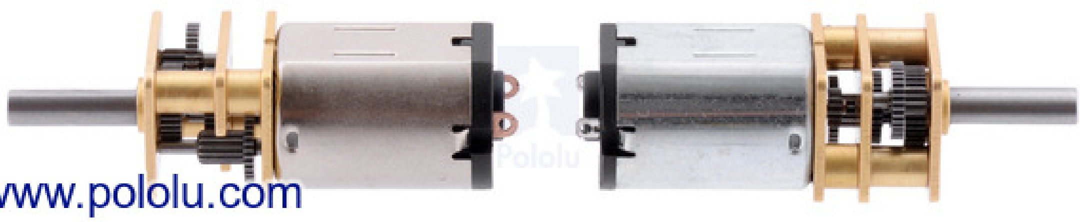
Rated Voltage	Stall Current	No-Load Current	No-Load Speed (RPM)	Extrapolated Stall Torque		Max Power (W)	Approx Gear Ratio	No Encoder	w/ Extended Motor Shaft	w/ Encoder, Back Conn.	w/ Encoder, Side Conn.
				(kg·cm)	(oz·in)						
LP 6V (low-power)											
6 V	0.36 A	50 mA	2500	0.05	0.7	–	5:1	#1100	#2200	#5100	#5101
		40 mA	1300	0.10	1.4	–	10:1	#1099	#2201	#5102	#5103
			860	0.17	2.4	0.37	15:1	#4780	#4781	#5104	#5105
			450	0.29	4.0	0.31	30:1	#993	#2202	#5106	#5107
			270	0.44	6.1	0.29	50:1	#1098	#2203	#5108	#5109
			180	0.64	8.9	0.29	75:1	#2360	#2209	#5110	#5111
			130	0.74	10	0.25	100:1	#992	#2204	#5112	#5113
			90	1.1	15	0.25	150:1	#1097	#2205	#5114	#5115
			65	1.6	22	0.25	210:1	#1096	#2206	#5116	#5117
			54	1.7	24	0.23	250:1	#1095	#2207	#5118	#5119
			45	2.0	28	0.22	298:1	#1094	#2208	#5120	#5121
			36	2.9	40	0.27	380:1	#4790	#4791	#5122	#5123
			13	5.5	76	–	1000:1	#1596	#3058	#5124	#5125

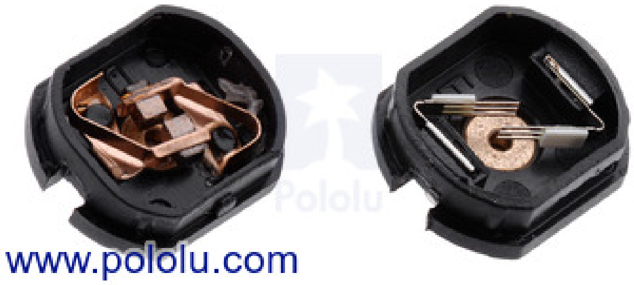
Note: Stalling or overloading gearmotors can greatly decrease their lifetimes and even result in immediate damage. The recommended upper limit for instantaneous torque is 2.5 kg·cm (35 oz·in) for the 380:1 and 1000:1 gearboxes, and 2 kg·cm (25 oz·in) for all the other gear ratios; we strongly advise keeping applied loads **well under** this limit. Stalls can also result in rapid (potentially on the order of seconds) thermal damage to the motor windings and brushes, especially for the versions that use high-power (HP and HPCB) motors; a general recommendation for brushed DC motor operation is 25% or less of the stall current.

In general, these kinds of motors can run at voltages above and below their nominal voltages; lower voltages might not be practical, and higher voltages could start negatively affecting the life of the motor.

Motor windings/brush options

The 6V and 12V HPCB motors have long-life carbon brushes, and they offer the same performance at their respective nominal voltages, just with the 12 V motor drawing half the current of the 6 V motor. The 6V HP, MP, and LP motors have shorter-life precious metal brushes, which are generally lower-friction than carbon brushes and preferred for lower-current applications. The HPCB versions (shown on the left in the picture below) can be differentiated from versions with precious metal brushes (shown on the right) by their copper-colored terminals. Note that the HPCB terminals are 0.5 mm wider than those on the other Micro Metal Gearmotor versions (2 mm vs. 1.5 mm), and they are about 1 mm closer together (6 mm vs. 7 mm).



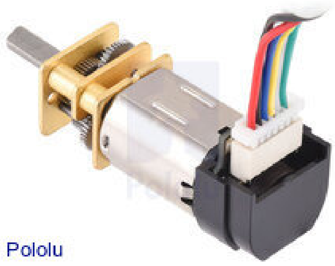


Micro Metal Gearmotor HPCB long-life carbon brushes (left) next to Micro Metal Gearmotor HP precious metal brushes (right).

Encoder options

Micro Metal Gearmotor (standard version without encoder or extended motor shaft).

Micro Metal Gearmotor with 12 CPR encoder, back connector (cable not included).



Micro Metal Gearmotor with 12 CPR encoder, side connector (cable not included).

Micro Metal Gearmotor with extended motor shaft.

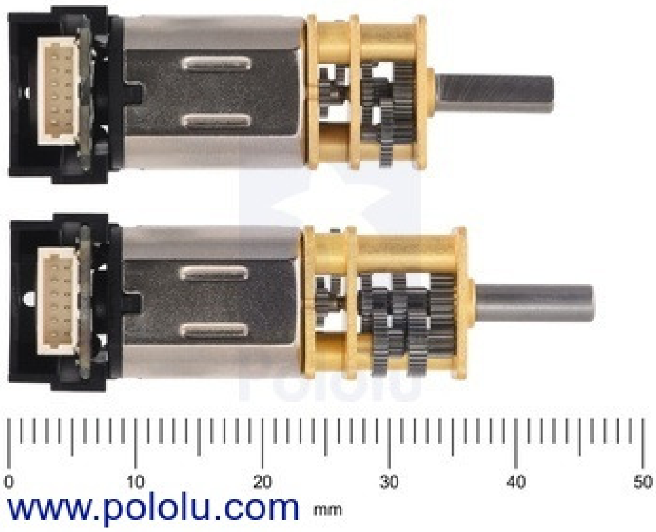
In addition to standard versions that are not intended for use with encoders, we have versions of our Micro Metal Gearmotors available with integrated 12 CPR quadrature encoders on the motor shafts (i.e. on the inputs to the gearbox). These are available in two styles—back connector and side connector—and work with our assortment of **6-pin JST SH-style cables** and **6-pin JST SH-style connector boards** (**cables are not included**). A plastic snap-on housing covers the encoder disc and electronics.

We also have motor versions available with an extended motor shaft for adding your own encoder. This 1 mm diameter shaft extends 4.5 mm from the rear of the motor and rotates at the same speed as the input to the gearbox. These versions work with our separately available **encoders for Micro Metal Gearmotors**.

Gearbox options

The table below shows the exact gear ratios for our Micro Metal Gearmotors. All gearboxes have the same overall dimensions except for 1000:1, which is 3.5 mm longer than the others. The gearboxes use brass plates on all versions except 380:1, which has steel plates for increased durability and resistance to wear from radial loads.

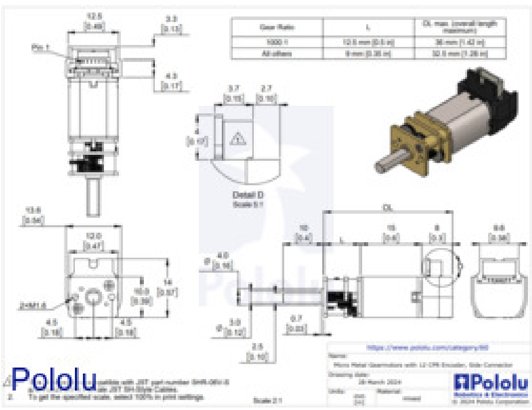
Nominal	Exact	Nominal	Exact
5 : 1	$\frac{27 \times 37}{20 \times 10} = 4.995 : 1$	150 : 1	$\frac{25 \times 32 \times 34 \times 35 \times 38}{12 \times 11 \times 14 \times 13 \times 10} \approx 150.5828 : 1$
10 : 1	$\frac{35 \times 37}{13 \times 10} \approx 9.9615 : 1$	210 : 1	$\frac{25 \times 34 \times 34 \times 35 \times 38}{12 \times 9 \times 13 \times 13 \times 10} \approx 210.5906 : 1$
15 : 1	$\frac{25 \times 34 \times 31}{12 \times 9 \times 16} \approx 15.2488 : 1$	250 : 1	$\frac{25 \times 34 \times 37 \times 35 \times 38}{12 \times 10 \times 10 \times 14 \times 10} \approx 248.9792 : 1$
30 : 1	$\frac{31 \times 33 \times 35 \times 34}{16 \times 14 \times 13 \times 14} \approx 29.8609 : 1$	298 : 1	$\frac{25 \times 34 \times 37 \times 35 \times 38}{12 \times 9 \times 10 \times 13 \times 10} \approx 297.9238 : 1$
50 : 1	$\frac{32 \times 33 \times 35 \times 38}{15 \times 14 \times 13 \times 10} \approx 51.4462 : 1$	380 : 1	$\frac{25 \times 35 \times 39 \times 36 \times 39}{12 \times 9 \times 9 \times 13 \times 10} = 379.1\bar{6} : 1$
75 : 1	$\frac{34 \times 34 \times 35 \times 38}{13 \times 12 \times 13 \times 10} \approx 75.8126 : 1$	1000 : 1	$\frac{25 \times 34 \times 35 \times 34 \times 34 \times 34 \times 27}{12 \times 9 \times 12 \times 14 \times 14 \times 14 \times 9} \approx 986.4064 : 1$
100 : 1	$\frac{35 \times 37 \times 35 \times 38}{12 \times 11 \times 13 \times 10} \approx 100.3700 : 1$		



Micro Metal Gearmotor size comparison of 1000:1 (bottom) vs other gear ratios (top).

Please note that the higher gear ratios can generate enough torque to damage themselves if exposed to loads beyond what they are rated for (25 kg·mm for 1000:1 and 380:1, 20 kg·mm for everything else). The point of these higher gear ratios is not to deliver more overall torque but rather to allow for slower speeds at a given voltage and to draw less current for a given torque within its rated operating range.

Dimensions



Our Micro Metal Gearmotors all have gearboxes with a 10 × 12 mm cross section and a 9 mm-long, 3 mm-diameter D-shaped gearbox output shaft. The gearbox length is the same for all gear ratios except 1000:1, which is 3.5 mm longer than the others. On versions without an encoder, the motor fits entirely within the 10 × 12 mm gearbox cross section. The two encoder options fit within the gearbox cross section on the side opposite the connector and extend past it by 3 mm or 4 mm on the connector side, depending on the connector orientation; for the other two sides, the encoder extends past the gearbox cross section by 0.25 mm.

In terms of size, these gearmotors are very similar to Sanyo's popular 12 mm NA4S DC gearmotors, and gearmotors with this form factor are sometimes also referred to as N20 motors. The versions with carbon brushes (HPCB) have slightly different terminal and end-cap dimensions than the versions with precious metal brushes, but all of the other dimensions are identical.

See our detailed **dimension diagrams** (635k pdf) for more information.

Details for item #3073

Gearmotor details:

motor/brush type	gearbox	encoder
HPCB 6V: high-power 6V with carbon brushes	51.45:1 with brass plates	encoder-compatible (extended motor shaft)

Micro Metal Gearmotor HPCB with extended motor shaft.

Key specifications:

voltage	no-load performance	stall extrapolation*
6 V	650 RPM, 150 mA	0.74 kg·cm (10 oz·in), 1.5 A

* Note: Stall torque and stall current specifications are theoretical values; stalls could damage the motor or gearbox.

Exact gear ratio: $\frac{32 \times 33 \times 35 \times 38}{15 \times 14 \times 13 \times 10} \approx \mathbf{51.45:1}$

Motor Accessories

- **Wheels and hubs:** The Micro Metal Gearmotor's output shaft matches our assortment of **Pololu wheels for 3mm shafts** and the **Solarbotics RW2i rubber wheel**. You can also use our **Pololu universal mounting hubs** to mount custom wheels and mechanism to the Micro Metal Gearmotor's output shaft, and you can use our **12mm hex wheel adapter** to use this motor with many common hobby RC wheels.

Pololu Wheel 32×7mm on a Micro Metal Gearmotor.

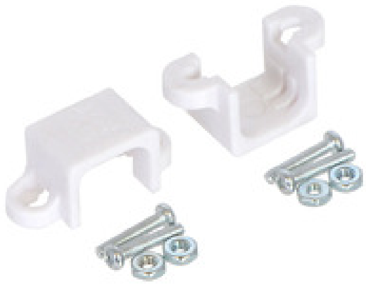
Black Pololu Wheel 70×8mm on a Pololu Micro Metal Gearmotor.

A pair of Pololu universal aluminum mounting hubs for 3 mm diameter shafts.

12mm Hex Wheel Adapter for 3mm Shaft on a Micro Metal Gearmotor.

- **Mounting brackets:** Our **mounting bracket** (also available in **white**) and **extended mounting bracket** are specifically designed to securely mount the gearmotor while enclosing the exposed gears. We recommend the **extended** mounting bracket for wheels with recessed hubs, such as the Pololu wheel 42×19mm. Our Micro Metal Gearmotors will also work with our **15.5D mm metal gearmotor bracket pair**.

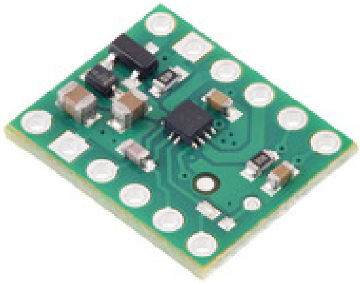
Black micro metal gearmotor mounting bracket pair with included screws and nuts.



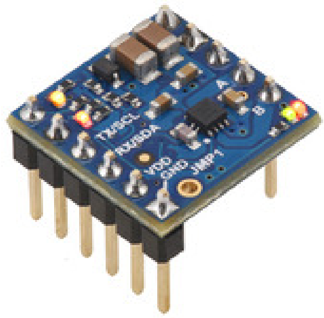
White micro metal gearmotor mounting bracket pair with included screws and nuts.

Pololu micro metal gearmotor bracket extended with Micro Metal Gearmotor.

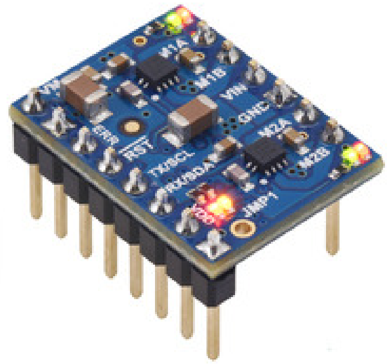
- **Motor controllers and drivers:** We have a number of **motor controllers** and **motor drivers** that can be used to control these Micro Metal Gearmotors. In particular, the **MP6550 motor driver** is well suited for these motors, as are our Motoron **M1x550**, **M2x550**, **M3S550**, and **M3H550** chainable serial motor controllers.



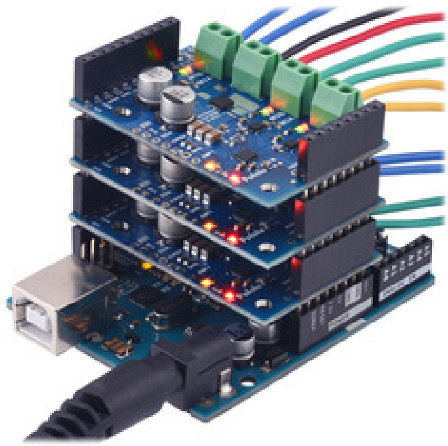
MP6550 Single Brushed DC Motor Driver Carrier.



Motoron M1T550/M1U550 Single Motor Controller (Header Pins Soldered).



Motoron M2T550/M2U550 Dual Motor Controller (Header Pins Soldered).



Three Motoron M3S550 shields being controlled by an Arduino Uno.

- **Current sensors:** The recommended motor control options above have built-in current sensing, but we also have an assortment of stand-alone Hall effect-based **current sensors** to choose from:

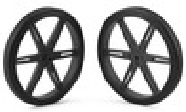
ACS724LLCTR-2P5AB Current Sensor Carrier -2.5A to +2.5A.

We also incorporate these motors into some of our products, including our **Zumo robot** and **3pi robot** :

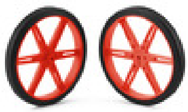
Assembled Zumo 2040 robot.

3pi+ 2040 Robot.

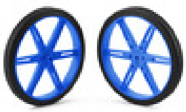
People often buy this product together with:



[Pololu Wheel 80×10mm Pair - Black]



[Pololu Wheel 80×10mm Pair - Red]



[Pololu Wheel 80×10mm Pair - Blue]